deduced that black holes are ‘hot’. They behave like hot bodies: they emit heat. In doing so, they lose energy and hence mass (since energy and mass are the same thing), becoming progressively smaller. They ‘evaporate’. This ‘evaporation of black holes’ is the most important discovery made by Hawking.

Objects are hot because their microscopic constituents move. A hot piece of iron, for example, is a piece of iron where the atoms vibrate very rapidly around their equilibrium position. Hot air is air in which molecules move faster than in cold air.

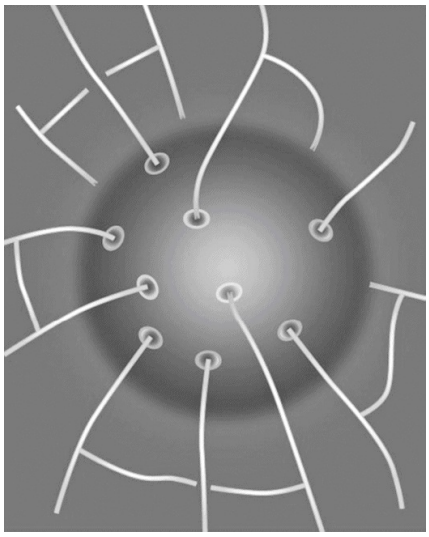


Figure 10.2 The surface of a black hole crossed by loops, that is to say, by links of the spin network that describe the state of the gravitational field. Each loop corresponds to a quantum area of the black hole’s surface. © John Baez.

What are the elementary ‘atoms’ that vibrate, making a black hole hot? Hawking left this problem unanswered. Loop theory provides a possible answer. The elementary atoms of a black hole that vibrate, and are thus responsible for its temperature, are the individual quanta of space on its surface.

Thus, it is possible to understand the peculiar heat of black holes predicted by Hawking using loop theory: the heat is the result of the microscopic vibrations of the individual atoms of space. These vibrate because in the world of quantum mechanics *everything* vibrates; nothing stays still. The impossibility of anything being entirely and continuously still in a place is at the heart of quantum mechanics. Black-hole heat is directly connected to loop quantum gravity’s fluctuations of the atoms of space. The precise position of the black hole’s horizon is determined only in relation to these microscopic fluctuations of the gravitational field. Hence, in a certain sense, the horizon fluctuates like a hot body.